NEEL BANSAL · ANDERS HAROLDSON



HACKHARVARD 2026 · SPACEXAI CHALLENGE

TEARDOWN

BRINGING ROCKET SCIENCE DOWN TO EARTH

THE PROBLEM

Kids watch rockets launch. They never get to fly one.

The facts are public, but they are scattered, and a webcast only lets you watch.

> So kids never feel how one lean changes where a rocket ends up.



NO WAY TO TOUCH IT

Spread over three sites in three formats, and you can't hold any of it.

THE TEAM

NEEL AND ANDERS + CURSOR AND GROK

The two of us built every part of it together, side by side.

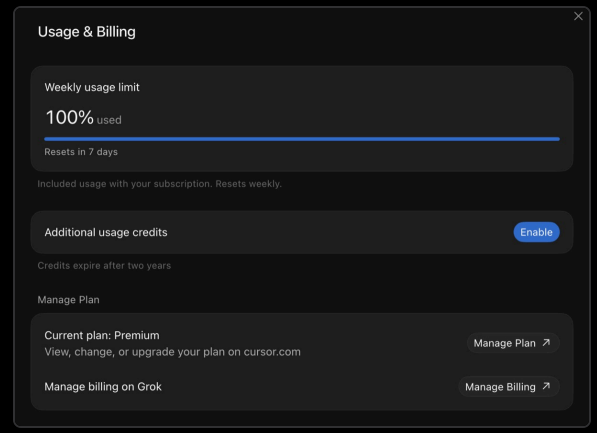
> We used Cursor and Grok as hard as we could. That is our real usage.

NEEL BANSAL + ANDERS HAROLDSON

Built together, start to finish.

CURSOR + GROK

Used as far as they would go.



Our real Grok Bot usage during the hackathon.

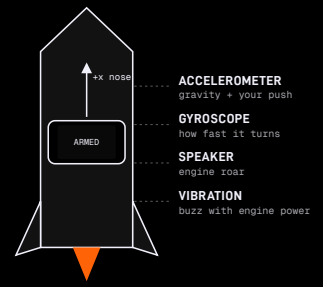
HARDWARE FOR GOOD

YOUR PHONE IS THE ROCKET

We planned to solder our own sensor board, but Micro Center and the other stores nearby didn't have the parts. Every phone already has them.

So a phone slides into a cardboard rocket and becomes its motion sensor, its speaker and its rumble. It runs in a web page, so there is no app to install.

> We checked with the HackHarvard co-director: the phone's accelerometer and speaker count as hardware for this track.



MOVE	WHAT THE SENSOR MUST SEE	DOES
LIFT	≥ 2.5 m/s² up the nose for 180 ms	Launch
LEAN	calibrated tilt past a 2° dead zone, ≤ 12°/s	Steer
PUSH	≥ 3 m/s² along the nose	Engine power
FLICK	≥ 12 m/s² sideways for 40 to 220 ms	Drop stage 1
NOSE DOWN	past 150° for 2 s	Abort
SILENCE	no frame for 2 s	Pause

LIVE DEMO

ONE RUN, START TO FINISH

Explode the rocket and push engine power. The cardboard rocket in your hands roars and shakes.

Calibrate, launch, lean to steer. At 8× the Earth curves away in under a minute.

Reach orbit, or crash. Either way Grok Imagine draws a postcard of your flight to download.

1 TAKE IT APART

2 FLY IT

3 KEEP THE POSTCARD

Backup if the phone won't pair: npm run sim:phone.

UNDER THE HOOD

ONE STORE, THREE WAYS IN

The phone, the mouse and Grok all drive one store, so Grok uses the same actions as the buttons.

Real orbital physics, 60 steps a second. 1,116 unit tests, plus a flight test with a simulated phone.



SPACEXAI CHALLENGE

BUILT FOR THIS TRACK

The running app meets every requirement, and you just saw each one in the demo.

Made by Neel Bansal and Anders Haroldson

Student project. Not affiliated with SpaceX or xAI.

Model: “Falcon 9 - SpaceX” by Stanley

Creative, CC BY 4.0. Model page

✓ **REAL SPACE DATA**	Falcon 9 specs from spacex.com, Sentinel-6 published milestones, NASA Earth imagery, a sourced emissions file
✓ **GROK VOICE OR IMAGINE**	Both: Grok Voice narrates every part and talks live, Imagine draws a postcard of every flight, plus function calling
✓ **BUILT WITH CURSOR**	Built in Cursor, plans in docs/plans/, using Grok Bot
✓ **HARDWARE FOR GOOD**	A phone in a cardboard rocket: accelerometer, gyro, speaker, vibration. Confirmed with the co-director

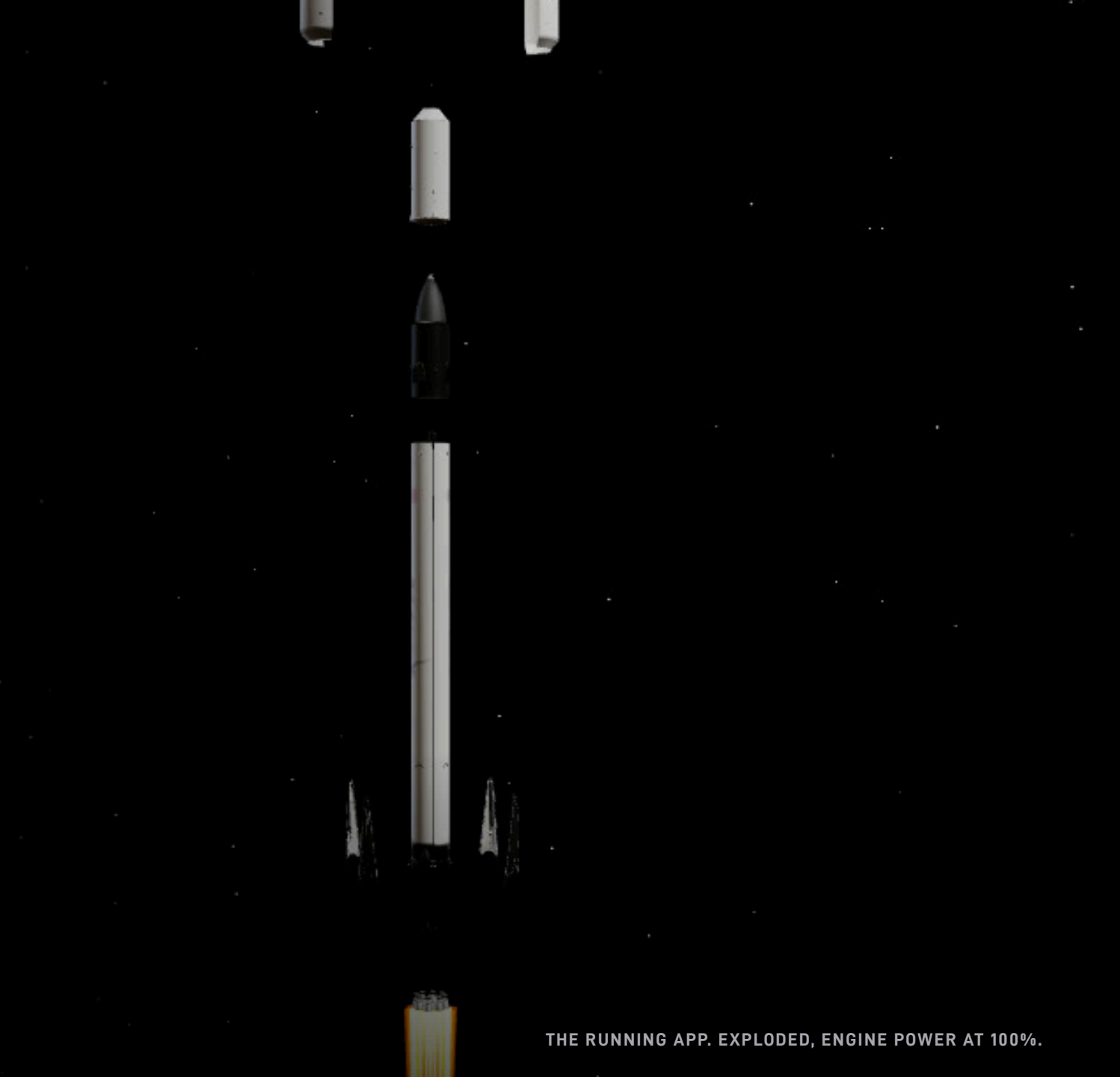
TAKE IT APART

Slide Explode and the Falcon 9 splits into its 10 real parts. Tap one for a card with real specs and a source link.

> Push Engine power and nine Merlin plumes grow. The phone inside the cardboard rocket roars and vibrates with them, so a kid feels every burn in the rocket they are holding.

Press Start tour and Grok Voice flies to each part and explains it out loud, at Kid, Curious or Engineer level. Press Voice to talk back live.

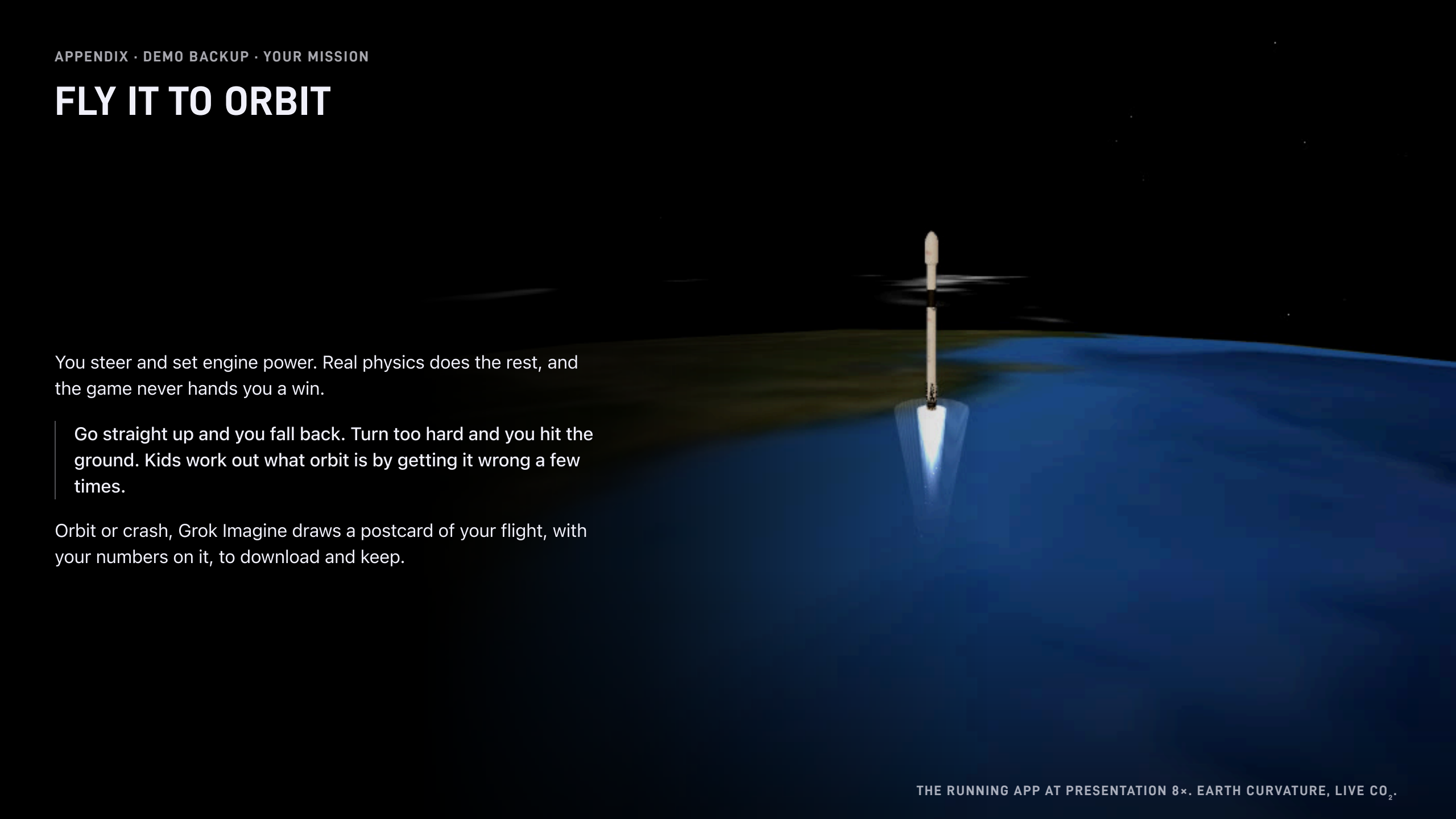
Or type or tap a question chip: Grok selects parts, explodes the rocket and flies the camera with the same actions the buttons use.



THE RUNNING APP. EXPLODED, ENGINE POWER AT 100%.

APPENDIX · DEMO BACKUP · YOUR MISSION

FLY IT TO ORBIT



You steer and set engine power. Real physics does the rest, and the game never hands you a win.

> Go straight up and you fall back. Turn too hard and you hit the ground. Kids work out what orbit is by getting it wrong a few times.

Orbit or crash, Grok Imagine draws a postcard of your flight, with your numbers on it, to download and keep.

THE RUNNING APP AT PRESENTATION 8×. EARTH CURVATURE, LIVE CO_2.